Exam Date & Time: 23-Feb-2026 (01:15 PM - 03:30 PM)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Computer Science and Applications
University Theory Examination February 2026
1st Semester of Master of Science in Information Technology (M.Sc.(IT))

WEB DEVELOPMENT USING .NET [CTUC506]

Marks: 50 **Duration: 135 mins.**

I

Answer all the questions. Section Duration: 20 mins

1 Which feature of .NET Framework would you argue is most criticalfor cross-language integration?

1) CLS 2) CLR 3) FCL 4) CTS (1)

Course Outcome: "CO1" / Cognitive Level: "Application"

2 How does managed code differ from unmanaged code in terms ofexecution environment?

1) Managed code runs without CLR 2) Managed code relies on CLR services 3) Unmanaged code uses namespaces 4) Unmanaged code supports MSIL (1)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

3 Which control is most effective for selecting multiple options?

1) RadioButtonList 2) Textbox 3) DropDownList 4) Label (1)

Course Outcome: "CO1" / Cognitive Level: "Evaluate"

4 Which ADO.NET component is most critical for disconnectedarchitecture?

1) SqlCommand 2) DataReader 3) DataSet 4) SqlConnection (1)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

5 Which method of SqlCommand executes queries returning a singlevalue?

1) ExecuteReader 2) ExecuteScalar 3) ExecuteNonQuery 4) ExecuteXmlReader (1)

Course Outcome: "CO2" / Cognitive Level: "Application"

6 Which component in MVC handles user input?

1) Model 2) Controller 3) Razor 4) None of the above (1)

Course Outcome: "CO3" / Cognitive Level: "Application"

7 Which protocol does SOAP rely on for message formatting?

1) UDP 2) TCP 3) XML 4) None of the above (1)

Course Outcome: "CO4" / Cognitive Level: "Application"

8 Which REST method is used to update existing resources?

1)	PUT		2)	POST		3)	DELETE		4)	GET	

(1)

Course Outcome: "CO4" / Cognitive Level: "Application"

9 Which control is essential for managing client-side scripts in anASP.NET AJAX application?

1)	UpdatePanel		2)	ScriptManager		3)	Timer		4)	None of the above	

(1)

Course Outcome: "CO5" / Cognitive Level: "Application"

10 Which view engine is the default in ASP.NET MVC?

1)	Web Forms View Engine		2)	Razor View Engine		3)	XAML View Engine		4)	HTML View Engine	

(1)

Course Outcome: "CO3" / Cognitive Level: "Remember"

II

Answer 4 out of 6 questions.

1 How do the structural and functional differences between ASP andASP.NET influence the way developers build, maintain, and scale webapplications?

(5)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

2 Write down the difference between DataReader and Dataset.

(5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

3 Define JIT, draw the execution diagram and explain it with itstypes in detail.

(5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

4 What is the use of ADO.NET Data Provider? Explain each component indetails with an example.

(5)

Course Outcome: "CO2" / Cognitive Level: "Application"

5 Write down the code to find the factorial of the given number. Thenumber is taken from the user.

(5)

Course Outcome: "CO1" / Cognitive Level: "Create"

6 Write a code to perform insert and delete operations on the CourseTable. Assume any four columns of the course table.

(5)

Course Outcome: "CO2" / Cognitive Level: "Create"

III

Answer 4 out of 6 questions.

1 What is the use of HTML Helper methods? Explain any two helpermethods along with an example.

(5)

Course Outcome: "CO3" / Cognitive Level: "Application"

2		

Write down the difference between ASP.NET and MVC.NET.

(5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

3		

Write down the difference between .NET Framework and .NET CoreFramework.

(5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

4		

Explain ServiceContract, DataContract, and OperationContract alongwith an example.

(5)

Course Outcome: "CO4" / Cognitive Level: "Application"

5		

What is the use of JSON? Create an array of any 10 objects in whichthe structure of each object remains the same.

(5)

Course Outcome: "CO4" / Cognitive Level: "Application"

6		

Explain the features of the .NET Core framework.

(5)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
University Exam Dec-Feb 2026	CTUC506:WEB DEVELOPMENT USING .NET

#	Course Outcome	Description	Marks
1	CO1	Students will be able to understand .NET framework and develop ASP.NET application.	18
2	CO2	Students will be able to develop database centric applications.	17
3	CO3	Students will be familiar with concepts of MVC for application development.	12
4	CO4	Students will able to understand and develop SOA based applications with .NET framework.	12
5	CO5	Students will be familiar with the concepts of AJAX	11
		Total	**70**

* NA - CO Not Selected

Event Name	Subject
University Exam Dec-Feb 2026	CTUC506:WEB DEVELOPMENT USING .NET

#	Level	No of Questions	Marks (%)
1	Remember	1	1(1.43%)
2	Understand	2	10(14.29%)
3	Application	10	26(37.14%)
4	Analysis	5	21(30%)
5	Create	2	10(14.29%)
6	Evaluate	2	2(2.86%)
	Total	22	70 (100%)